

Hand Detection & 21-Keypoint Model

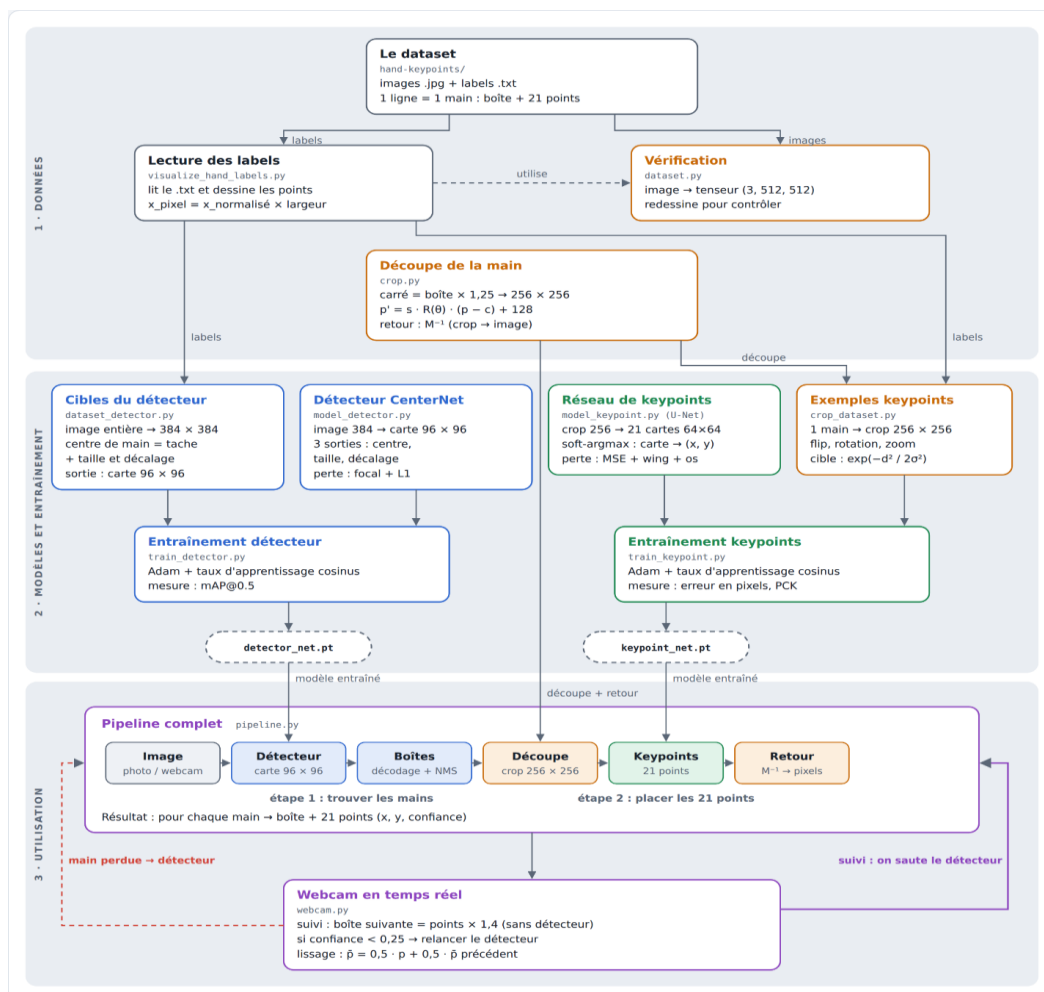
Project journal · summary

1. Objective

Build, **from scratch in PyTorch**, a real-time system that detects hands and places **21 keypoints** on each hand from a webcam. The goal is touchless human-computer interaction: gesture commands, sign-to-text for people with hearing or speech disabilities, air writing and laptop control.

Item	Detail
Dataset	Ultralytics hand-keypoints: 18,776 train / 7,992 val hands, 68 values per hand (box + $21 \times (x, y, v)$)
Hardware	Intel Core i7 11th Gen, NVIDIA RTX 3060, 16 GB RAM
Approach	Two stages, like MediaPipe Hands: detector → crop → keypoint network
Improvement loop	Human-in-the-loop: user corrects boxes and points live, then the model is fine-tuned

2 Model pipeline

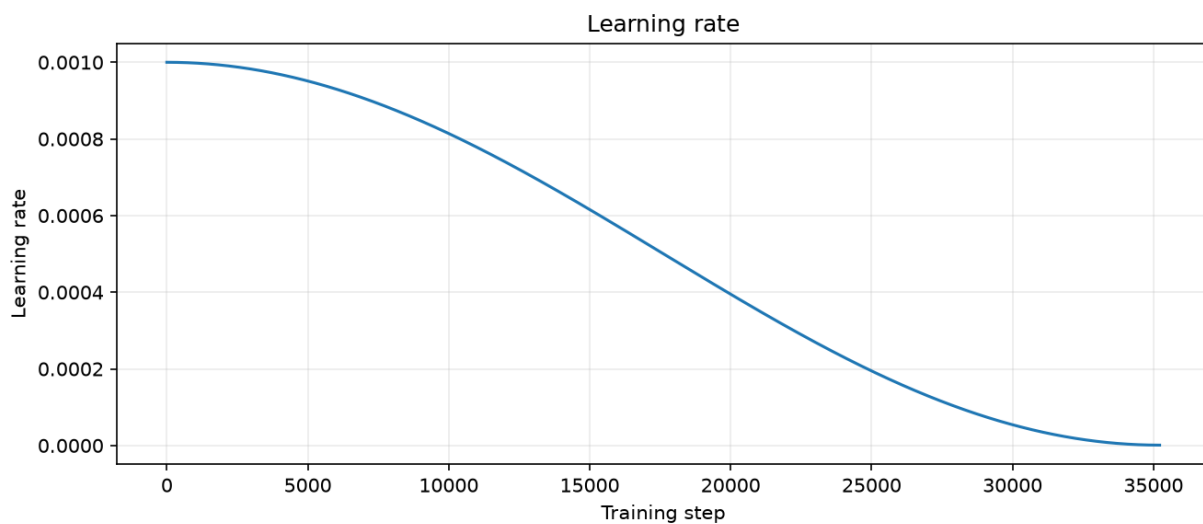


Stage	Model	Input → output	Key idea
1 · Find hands	CenterNet (5.96 M)	384×384 image → 96×96 maps	hand centre = Gaussian peak, plus size and offset
2 · Crop	Affine transform M	$\text{box} \times 1.25 \rightarrow 256 \times 256$ crop	same M on image and points; M^{-1} to go back
3 · Place points	U-Net (3.49 M)	crop → 21 heatmaps 64×64	soft-argmax turns each heatmap into (x, y)
4 · Real time	Tracking + smoothing	points → next box	detector runs only when the hand is lost

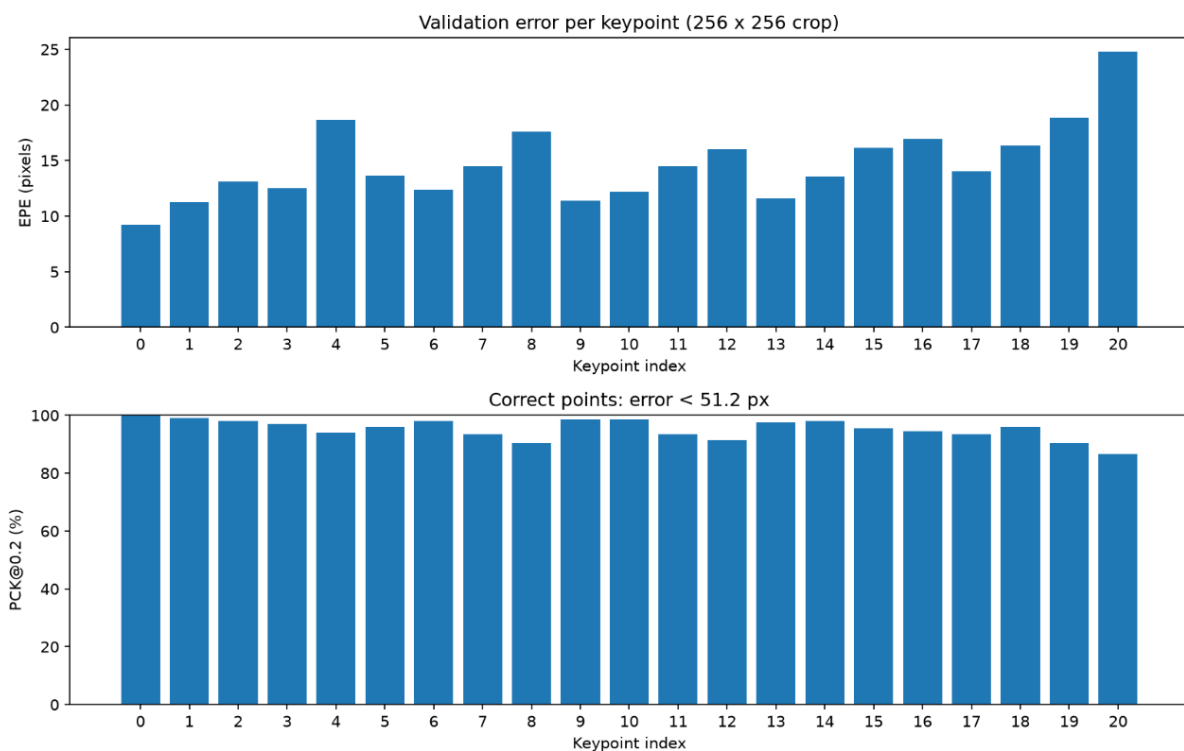
3. Key mathematics

Where	Function
Crop	$p' = s \cdot R(\theta) \cdot (p - c) + 128, \quad s = 256 / (1.25 \cdot \max(w, h))$
Heatmap target	$H(x, y) = \exp(-[(x - u)^2 + (y - v)^2] / 2\sigma^2)$
Detector loss	$L = \text{focal}(\text{heatmap}) + 0.1 \cdot L1(\text{size}) + L1(\text{offset})$
Keypoint loss	$L = \text{MSE}(\text{heatmap}) + 0.1 \cdot \text{wing} + 0.05 \cdot \text{bone}$
Soft-argmax	$u = \sum x \cdot \text{softmax}(\beta \cdot H)$
Optimiser	Adam, cosine learning rate $1e-3 \rightarrow 0$
Box decoding	peak = $\text{maxpool}_{3 \times 3}$, then NMS with IoU > 0.5
Webcam	$\bar{p}_t = 0.5 \cdot p_t + 0.5 \cdot \bar{p}_{t-1}$

4. Training results (30 epochs)



Model	Metric	Result	Reading
Detector	Precision	99 % (98 / 99)	almost no false alarms
Detector	Recall	98 % (98 / 100)	misses 2 hands in 100
Detector	Train / val loss	2.0 / 2.45	small gap, no overfitting
Keypoints	EPE	≈ 14.7 px (from 48)	≈ 7 % of hand size
Keypoints	PCK@0.2	≈ 95 % (from 61 %)	95 of 100 points within 51 px
Keypoints	Per point	wrist ≈ 9 px, little tip ≈ 25 px	error grows toward fingertips
Both	Learning rate	cosine $1e-3 \rightarrow 0$	as planned



5. Findings

Works. Both models converge smoothly with full-dataset epochs, and validation follows training without diverging.

Loss balance. The detector loss is dominated by the box-size term (pixels); the keypoint loss is dominated by the wing term, while the heatmap term is almost zero.

Weak spot. Fingertips, especially the little finger. A soft-argmax grid mismatch (63 vs 64 intervals) can add up to ≈ 4 px near crop edges.

Evaluation limits. The detector was scored on 100 hands only, and keypoint scores use ground-truth crops, not detector crops. Webcam performance is not measured yet.

6. Next steps

1. Evaluate on the full validation set (mAP@0.5 and mAP@0.75, score threshold 0.01).

2. Fix the soft-argmax grid ($\text{arange}(64) / 64$) and lower β to 20–50.
3. Rebalance losses: heatmap $\times 100$; detector size in grid units or log scale.
4. Measure end-to-end keypoint error on detector crops, then test on the live webcam.
5. Start the human-in-the-loop cycle: collect corrected samples and fine-tune.

Prepared by

Maroune Morfi

Maroune Morfi · 28 September 2026